

EXPERIMENT 11

11 Arranging the Metals According to the Decreasing Order of their Reactivity

Aim

To study the reactions of the metals Al, Zn, Fe and Cu with the solutions of salts $\text{Al}_2(\text{SO}_4)_3$, ZnSO_4 , FeSO_4 , CuSO_4 and to arrange these metals in decreasing order of their reactivity.

Requirements

Test tubes, test tube stand, distilled water, sandpaper, forceps, the metals Aluminium, Zinc, Iron and Copper, and solutions of the salts $\text{Al}_2(\text{SO}_4)_3$, ZnSO_4 , FeSO_4 , CuSO_4 .

Procedure

1. Take four 50 ml beakers and label them as Aluminium sulphate, Zinc sulphate, Ferrous sulphate and Copper sulphate.
2. Clean all the metallic pieces using sandpaper and cut them into small pieces.
3. Pour 10 ml each of the $\text{Al}_2(\text{SO}_4)_3$, ZnSO_4 , FeSO_4 and CuSO_4 solutions into their respective labelled beakers.
4. Drop two pieces of aluminium metal into the beakers containing ZnSO_4 , FeSO_4 and CuSO_4 solutions.
5. Leave the beakers undisturbed for about 20 minutes.
6. After about 20 minutes, note any change in the colour of the solutions or any other change.
7. Repeat the same procedure by adding: Zinc pieces to the $\text{Al}_2(\text{SO}_4)_3$, FeSO_4 and CuSO_4 solutions; Iron pieces to the $\text{Al}_2(\text{SO}_4)_3$, ZnSO_4 and CuSO_4 solutions; Copper pieces to the $\text{Al}_2(\text{SO}_4)_3$, ZnSO_4 and FeSO_4 solutions.
8. Record your observations.

Figure

(Draw and label the diagrams of metals with the solutions of salts)



Pieces of aluminium metal strip



Pieces of zinc metal strip



Pieces of iron metal strip



Pieces of copper metal strip



Aluminium sulphate solution



Zinc sulphate solution



Ferrous sulphate solution



Copper sulphate solution

Observation Table

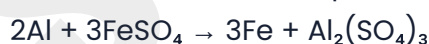
Metal	$\text{Al}_2(\text{SO}_4)_3$	ZnSO_4	FeSO_4	CuSO_4
Aluminium	—	Solution remains colourless. Zinc metal settles down at the bottom of the beaker. Zinc is displaced.	Green colour of FeSO_4 disappears. Iron metal settles down at the bottom of the beaker. Iron is displaced.	Blue colour of CuSO_4 disappears. The brown coloured copper particles settle down at the bottom of the beaker. Copper is displaced.
Zinc	(Solution remains colourless.) No reaction	—	Green colour of ferrous sulphate disappears. Iron metal settles down at the bottom of the beaker. Iron is displaced.	Blue colour of copper sulphate disappears and red-brown Copper particles settle down at the bottom of the beaker. Copper is displaced.
Iron	(Solution remains colourless.) No reaction	(Solution remains colourless) No reaction	—	Blue colour of Copper sulphate changes to light green coloured FeSO_4 . Copper metal is formed in the beaker. Copper is displaced.
Copper	(Solution remains colourless.) No reaction	(Solution remains colourless.) No reaction	(Solution remains colourless.) No reaction	—

Chemical Reactions

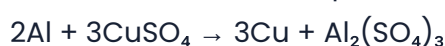
(1) More reactive Al displaces the less reactive Zn from ZnSO_4 solution.



(2) More reactive Al displaces the less reactive Fe from FeSO_4 solution.



(3) More reactive Al displaces the less reactive Cu from CuSO_4 solution.

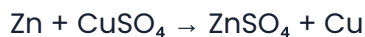


(4) Zn cannot displace Al from $\text{Al}_2(\text{SO}_4)_3$ solution.

(5) More reactive Zn displaces the less reactive Fe from FeSO_4 solution.



(6) More reactive Zn displaces the less reactive Cu from CuSO_4 solution.



(7) Fe cannot displace Al from $\text{Al}_2(\text{SO}_4)_3$ solution.

(8) Fe cannot displace Zn from ZnSO_4 solution.

(9) More reactive Fe displaces the less reactive Cu from CuSO_4 solution.



(10) Cu cannot displace Zn from ZnSO_4 solution, Cu cannot displace Al from $\text{Al}_2(\text{SO}_4)_3$ solution, Cu cannot displace Fe from FeSO_4 solution.

Inferences

1. Aluminium displaces Zn, Cu and Fe from their salt solutions.
2. Zinc displaces Cu and Fe from their salt solutions.
3. Fe displaces Cu from its salt solution.
4. Cu cannot displace Zn, Al or Fe from their salt solutions.
5. Thus, aluminium is more reactive metal.

The decreasing order of reactivity of metal is:



Most reactive -----> Least reactive
(Reactivity in decreasing order)

Precautions

1. Do not touch any chemical.
2. Label the beakers properly and put them in sequence. It is better to label the beakers both with the chemical names and also by chemical symbols.
3. It is very important that the apparatus you use must be very clean and dry.
4. Always clean the metallic strips before use with a sandpaper. Use the strips of metals of the same size, weight and number.

Multiple Choice Questions

1. To show that zinc is more reactive than copper, the correct procedure is to

- (A) prepare copper sulphate solution and dip zinc strip in it
- (B) prepare zinc sulphate solution and dip copper strip in it
- (C) heat together zinc and copper strips
- (D) add dil. nitric acid to both the strips

Answer: (A) prepare copper sulphate solution and dip zinc strip in it

2. A solution of $\text{Al}_2(\text{SO}_4)_3$ in water is not clear. It is due to

- (A) impurities present in water
- (B) hydrolysis of $\text{Al}_2(\text{SO}_4)_3$ in water
- (C) impurities present in $\text{Al}_2(\text{SO}_4)_3$
- (D) none of these

Answer: (D) none of these

3. Iron is

- (A) more reactive than zinc
- (B) more reactive than aluminium
- (C) less reactive than copper
- (D) less reactive than aluminium

Answer: (D) less reactive than aluminium

Remember

If a metal A displaces another metal B from the solution of its salt, then it means the metal A is more reactive than the metal B.

Date: _____

Teacher's Signature: _____